Polynomial-Time Exact Learning of Fourier-Sparse Boolean Functions from Nearly Linear Quantum Samples

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Abstract

We give an algorithm that exactly recovers the Fourier representation of every Boolean function with at most k nonzero Fourier coefficients, using $O(k \log^2(2k))$ independent uniform quantum examples and polynomial total computation and working space in the input dimension and k. Each quantum example is measured separately. The algorithm uses no membership queries to the target and no inverse of a state-preparation circuit. Fresh residual labels estimate the coefficients used to weight a subsequent independent batch of Fourier samples. A majority of signs of the resulting averages is an explicitly evaluable Boolean hypothesis. Classical heavy-Fourier search on this known hypothesis produces candidate indices, which a separate label batch certifies against the original target. Fresh certification protects the archive even when a proposal is inaccurate. A capped adaptive progress argument requires only a constant probability of contraction per trial, removing a logarithm from the previous version. We account for adaptive refitting, exact rational arithmetic, unsuccessful trials and all resource costs. This complements the smaller sample bound in the cited version of Ye by supplying a fixed polynomial computational guarantee. The bounds are asymptotic and do not assert practical running time.

1 Result, access model and contribution

Let $f : \mathbb{F}_2^n \to \{-1, 1\}$ and write

$$\chi_s(x) = (-1)^{s \cdot x}, \qquad c_s = \widehat{f}(s) = 2^{-n} \sum_x f(x)\chi_s(x), \qquad S = \operatorname{supp} \widehat{f}.$$

All norms and expectations on the cube use its uniform probability measure. The supplied quantum examples are independent copies of

$$|\psi_f\rangle = 2^{-n/2} \sum_x |x\rangle |b_f(x)\rangle, \qquad b_f(x) = (1 - f(x))/2. \tag{1}$$

The input consists only of these copies and integers n, k with $|S| \le k$. The cost of preparing the supplied examples is outside this access model; no preparation circuit or its inverse is supplied. A label batch means computational-basis measurements of fresh copies, giving independent uniform pairs $(X, f(X))$.

Theorem 1. *For $n \geq 1$ and $1 \leq k \leq 2^n$, every Boolean f with $|S| \leq k$ can be learned exactly with probability at least $2/3$ using*

$$O(k \log^2(2k))$$

supplied uniform quantum examples. The output is the complete list of pairs (s, c_s) with $c_s \neq 0$, with exact rational coefficients. Total quantum work is $O(nk \log^2(2k))$. Conservative classical bit-time and live-space bounds are

$$\widetilde{O}(n^2k^7 + nk^9) \quad \text{and} \quad \widetilde{O}(nk^2 + k^3), \tag{2}$$

respectively. Here only fixed powers of logarithms in n and k are suppressed. These resource caps hold on every measurement and random tape; an unsuccessful execution may return FAIL*. No chosen-input query to f is used.*

The theorem is about exact representation recovery, including coefficients that are rarely observed by Fourier sampling. It imposes no degree, Fourier-flatness or variance promise. The case $n = 0$ needs one label; an upper bound $k > 2^n$ can be replaced by 2^n.

Prior work and comparison. Arunachalam, Chakraborty, Lee, Paraashar and de Wolf [1] obtained an $O(k^{3/2} \log^2 k)$ quantum sample bound and a lower bound of $\Omega(k \log k)$ in their stated parameter range. Ye [2, version 1] improves the upper bound to $O(k \log^2 k / \log \log k)$ by constructing a grouped-sample candidate dictionary and proving support containment. That sample bound is smaller than ours. Ye's Theorem 1.1 states a sample guarantee; it does not state a fixed polynomial computation bound. Its Algorithms 1–2 explicitly construct and solve over the candidate dictionary. We do not assert a computational lower bound for that method. Caro et al. [3, Corollary 10] give an efficient exact learner with $O(k^4 \log(k^2/\delta))$ copies and success $1 - \delta$. Thus neither polynomial-time learning alone nor near-linear sample complexity alone is the contribution claimed here.

Our contribution is their simultaneous attainment through a computed approximation with error proportional to the current missing energy. We use Fourier sampling, Boolean granularity [4], concentration inequalities [6], and classical Fourier search [5] as attributed ingredients. The Fourier search is applied to a stored hypothesis, for which chosen-input evaluations are available. The target itself remains accessible only through supplied examples. We make no claim of optimal logarithmic factors or exhaustive publication priority.

Mechanism. The maintained approximation a has a certified support contained in S. Fresh labels estimate c_s at an error scale proportional to $\|f - a\|_2$. These estimates are fixed before drawing new Fourier samples. Thresholded reciprocals of the estimates then define an evaluable, bounded sign-majority hypothesis H. Searching H supplies candidate frequencies, including frequencies not directly sampled. Original-target certification and exact refitting protect the archive for every bounded proposal list; accurate proposals additionally contract the missing energy. Separating constant-confidence proposals from high-confidence archive safety permits a capped sequence of attempts, without requiring every proposal to succeed. Keeping $f - a$, rather than $f - H$, as the residual is essential: H can have a dense Fourier spectrum.

Revision scope. Version 0.1.0 used $O(k \log^3(2k))$ copies by assigning high confidence to every proposal. This revision uses constant proposal confidence and tolerates unsuccessful proposals while preserving archive safety, reducing the total to $O(k \log^2(2k))$. The target class, supplied-example model and polynomial bit-time and space bounds are unchanged.

2 Basic access and certificate lemmas

Parseval gives $\sum_s c_s^2 = 1$. We use the following consequence of Boolean granularity [4, Theorem 8.1]:

$$c_s \neq 0 \quad \Longrightarrow \quad |c_s| \geq 1/k. \tag{3}$$

For actual sparsity $s_0 > 1$, Theorem 8.1 of the cited April 27, 2011 author manuscript makes every coefficient an integer multiple of $2^{1-\lfloor \log_2 s_0 \rfloor}$. A nonzero coefficient therefore has magnitude at least $2/s_0 \geq 1/k$, proving (3). A sparsity-one Boolean function is a signed parity.

Apply Hadamard gates to all registers in (1). The amplitude of outcome $(s, 1)$ is $c_s/\sqrt{2}$. Thus the last bit is one with probability $1/2$, and conditional on that flag, s has probability c_s^2. Independent copies give independent flags and conditional samples. In particular, conditioning on enough flags in a fixed batch does not bias the successful indices.

For an archive $U \subseteq S$, put

$$W(U) = \sum_{s \in S \setminus U} c_s^2, \qquad P_U f = \sum_{s \in U} c_s \chi_s, \qquad \alpha^2 = 82/81. \tag{4}$$

The algorithm does not compute $W(U)$ or $P_U f$.

Lemma 2 (Fresh residual estimates)**.** *Let a be fixed before a fresh label batch, supported on at most k frequencies, and put $r = f - a$, $e = \|r\|_2$. Let $\mathcal{D}$ be any set of at most P frequencies fixed before that batch. For $0 < \varepsilon \leq 1$, $0 < \beta < 1$, take*

$$q \geq 32k\varepsilon^{-2} \ln \frac{4(P+1)}{\beta}. \tag{5}$$

Define $v = q^{-1} \sum_\ell r(X_\ell)^2$ and $z_s = q^{-1} \sum_\ell r(X_\ell)\chi_s(X_\ell)$. With probability at least $1 - \beta$,

$$e^2/2 \leq v \leq 3e^2/2, \qquad |z_s - \widehat{r}(s)| \leq \varepsilon e/\sqrt{k} \quad (s \in \mathcal{D}). \tag{6}$$

This conclusion also holds conditional on any history fixing $a, \mathcal{D}$ before the batch.

Proof. The residual has at most $2k$ Fourier terms, so $\|r\|_\infty \leq \sqrt{2k}e \leq \sqrt{3k}e$. Each $r(X)\chi_s(X)$ has variance at most e^2. Since $|\widehat{r}(s)| \leq e$, its centered absolute bound is at most $(\sqrt{2k}+1)e \leq 3\sqrt{k}e$. For the scalar bounded-case Bernstein inequality [6, Theorem 6.1(ii)], at deviation $t = \varepsilon e/\sqrt{k}$ the denominator satisfies

$$\sigma^2 + Bt/3 \leq e^2(1+\varepsilon) \leq 2e^2.$$

For q independent centered scalar variables bounded in absolute value by B, each with variance at most σ^2, that inequality bounds the deviation of their average by $2\exp[-qt^2/(2(\sigma^2 + Bt/3))]$. Thus a two-sided coefficient failure is at most $2\exp(-q\varepsilon^2/(4k))$. Also $\mathrm{Var}(r(X)^2) \leq 3ke^4$ and $|r(X)^2 - e^2| \leq 3ke^2$. At deviation $t = e^2/2$, the denominator is at most $2(3ke^4 + ke^4/2) = 7ke^4$, giving failure at most $2\exp(-q/(28k))$. Put $L_0 = \ln(4(P+1)/\beta)$. Equation (5) gives $q\varepsilon^2/(4k) \geq 8L_0$ and $q/(28k) \geq L_0$, so the union bound is at most $2(P+1)e^{-L_0} = \beta/2$. If $e = 0$, all residual values vanish and the conclusion is exact. □

Lemma 3 (Original-coefficient certification)**.** *Suppose $U \subseteq S$, a is supported on U, and a candidate list L, $|L| \leq P$, precedes a fresh batch satisfying* (5)*. Admit $s \in L \setminus U$ precisely when*

$$z_s^2 > 4\varepsilon^2 v/k. \tag{7}$$

On (6)*, no frequency outside S is admitted, and every proposed frequency with $|c_s| > 4\varepsilon e/\sqrt{k}$ is admitted.*

Proof. Outside U the mean of z_s is c_s. For $c_s = 0$, its squared estimate is at most $\varepsilon^2 e^2/k$, while the threshold is at least $2\varepsilon^2 e^2/k$. If $|c_s| > 4\varepsilon e/\sqrt{k}$, the estimated magnitude exceeds $3\varepsilon e/\sqrt{k}$, and the square root of the threshold is at most $\sqrt{6}\varepsilon e/\sqrt{k}$. The zero-residual case admits nothing. More generally, in any batch $v = 0$ forces every observed residual and every z_s to vanish, so no index is admitted even without the accuracy event. □

Lemma 4 (One Gram event controls adaptive refitting)**.** *For a fresh batch of*

$$q \geq 300(k+1)\ln\frac{2(k+1)}{\beta} \tag{8}$$

uniform labels, with probability at least $1-\beta$, every subset $U \subseteq S$ has a nonsingular empirical Gram matrix, and its exact least-squares fit a satisfies

$$\|P_U f - a\|_2 \leq \frac{\sqrt{W(U)}}{9}, \qquad \|f-a\|_2^2 \leq \alpha^2 W(U), \qquad \|a\|_2^2 \leq \alpha^2 < 2. \tag{9}$$

The set U may be chosen using this same batch. The empty fit is zero.

Proof. Let $S_0 = S \cup \{0\}$ and $p = |S_0| \leq k+1$. The analysis-only row $\phi(X) = (\chi_s(X))_{s\in S_0}$ satisfies $\|\phi(X)\|_2^2 = p$ and $\mathbb{E}\phi(X)\phi(X)^{\mathsf{T}} = I_p$. The independent positive semidefinite summands $\phi(X_\ell)\phi(X_\ell)^{\mathsf{T}}$ have largest eigenvalue p and their expected sum is qI_p. Matrix Chernoff [6, Corollary 5.2] therefore bounds the lower and upper eigenvalue tails by

$$p\left[\frac{e^{-\delta}}{(1-\delta)^{1-\delta}}\right]^{q/p}, \qquad p\left[\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right]^{q/p}, \quad 0<\delta<1,$$

respectively. The scalar exponents satisfy

$$\delta + (1-\delta)\ln(1-\delta) \geq \delta^2/2, \qquad (1+\delta)\ln(1+\delta) - \delta \geq \delta^2/3.$$

The first follows by integrating $-\ln(1-t) \geq t$; the second by integrating $\ln(1+t) \geq t - t^2/2$ on $[0,\delta]$. Summing the tails gives

$$\Pr\left[\left\|q^{-1}A_{S_0}^{\mathsf{T}}A_{S_0} - I_p\right\|_{\mathrm{op}} > \delta\right] \leq 2p\exp\left(-\frac{q\delta^2}{3p}\right), \quad 0<\delta<1.$$

Take $\delta = 1/10$; the budget in (8) is at least $300p\ln(2p/\beta)$ because both positive factors increase with p. This single event controls all principal submatrices and cross blocks, without a union bound over subsets. For $V = S \setminus U$, exact pointwise Fourier inversion gives the label vector $y = A_U c_U + A_V c_V$. The normal equations therefore give

$$a_U - c_U = (A_U^{\mathsf{T}}A_U/q)^{-1}(A_U^{\mathsf{T}}A_V/q)c_V.$$

The two operator norms are at most $10/9$ and $1/10$, respectively. This proves the first inequality; orthogonality proves the second. Finally

$$\|a\|_2 \leq \sqrt{1-W(U)} + \frac{\sqrt{W(U)}}{9} \leq \sqrt{1+1/81}.$$

The proof uses unknown S_0 only to define an event. The algorithm constructs only the columns it actually retains. If $U = S$, the fit is exactly f. □

3 Fourier search on a known Boolean function

We record the exact classical search interface needed below. It is the standard prefix-energy mechanism associated with Kushilevitz–Mansour [5]; the proof specifies the access and finite caps.

Lemma 5 (Capped heavy-spectrum search)**.** *Given deterministic evaluation access to $H : \mathbb{F}_2^n \to \{-1, 1\}$, a rational threshold $0 < \tau \le 1$, and $0 < \beta < 1$, let $P = \lceil 4/\tau^2 \rceil$. A classical algorithm returns at most P indices, or* FAIL*, using at most $2nP$ prefix tests and*

$$N = \left\lceil 32\tau^{-4} \ln \frac{4nP}{\beta} \right\rceil$$

samples per test. Each sample needs two evaluations of H and at most $2n$ random bits. With probability at least $1 - \beta$, it does not fail and its list contains every s with $|\widehat{H}(s)| \ge \tau$.

Proof. For a j-bit prefix u, orthogonality gives the bucket identity

$$\begin{aligned} E_u &= \sum_{v \in \mathbb{F}_2^{n-j}} \widehat{H}(u, v)^2 \\ &= \mathbb{E}_z \left(\mathbb{E}_x H(x, z)\chi_u(x)\right)^2 = \mathbb{E}_{z,x,y} H(x, z)H(y, z)\chi_u(x + y), \end{aligned} \tag{10}$$

where x, y are independent uniform j-bit strings, and the two evaluations share a uniform suffix z. This requires chosen inputs to H.

Start with the empty prefix. Test both children of each retained prefix with fresh samples of the last expression in (10); retain a child if its estimated energy is at least $3\tau^2/4$. Return FAIL immediately if the retained count exceeds P at any depth. Stop at depth n.

Each summand is in $[-1, 1]$, a range of length two. For one test, Hoeffding's average-deviation bound $2\exp(-Nt^2/2)$ at $t = \tau^2/4$ gives

$$\Pr[|\widetilde{E}_u - E_u| > \tau^2/4] \le 2\exp(-N\tau^4/32).$$

Condition on the past before each test and take a union bound over the tests. On simultaneous accuracy, every ancestor of a coefficient of magnitude at least τ survives. Every retained bucket has energy at least $\tau^2/2$; disjoint buckets at a fixed depth have total energy one. Hence at most $2/\tau^2 \le P$ survive and the cap does not fire. Fresh random inputs are generated classically and require no target examples. □

4 Constructing a queryable majority hypothesis

Assume $U \subseteq S$ and a fixed, known rational fit a satisfies (9). Write $W = W(U)$ and $e = \|f - a\|_2$. Draw an estimator label batch with Lemma 2, using list-size argument k in that lemma. For any index s, define

$$\widetilde{c}_s = \widehat{a}(s) + q_e^{-1} \sum_{\ell} (f(X_\ell) - a(X_\ell))\chi_s(X_\ell). \tag{11}$$

The formula is evaluated only at indices subsequently obtained from fresh Fourier measurements. Its values at all other indices need never be computed.

Conditional on the entire earlier history, S is fixed. Apply Lemma 2 to $\mathcal{D} = S$ in the analysis. This yields the event

$$e^2/2 \le v \le 3e^2/2, \qquad |\widetilde{c}_s - c_s| \le \Delta := \varepsilon e/\sqrt{k} \quad (s \in S), \tag{12}$$

with probability at least $1-\beta$. The union bound does not require knowing S. It is not a claim of uniform estimation over all 2^n indices.

After fixing the estimator batch, draw g independent groups of m Fourier samples s_{ij}, retaining multiplicities. A frequency passes the cutoff if $\widetilde{c}_s \neq 0$ and

$$\widetilde{c}_s^2 \geq 16\varepsilon^2 v/k. \tag{13}$$

Define $w_s = 1/\widetilde{c}_s$ on a pass and $w_s = 0$ otherwise. Form

$$Q_i(x) = \frac{1}{m}\sum_{j=1}^{m} w_{s_{ij}}\chi_{s_{ij}}(x), \qquad H(x) = \text{majority}_{1\leq i\leq g}\,\text{sign}\,Q_i(x), \tag{14}$$

where g is odd and $\text{sign}(0) = 1$. Zero estimates are never divided. This is an explicit finite classical circuit with exact rational weights. It has no call to f.

Lemma 6 (Bias and variance of the computed averages)**.** *Suppose $W > 0$ and* (12) *holds. Conditional on the estimator batch, let $h = \mathbb{E}Q_i$ as a function on the cube. Then*

$$\|h - f\|_2^2 \leq 36\varepsilon^2 e^2, \qquad \text{Var}(Q_i(x)) \leq \frac{9k}{4m} \quad \textit{for every } x. \tag{15}$$

Proof. The cutoff magnitude $t = 4\varepsilon\sqrt{v/k}$ obeys $2\sqrt{2}\Delta \leq t \leq 2\sqrt{6}\Delta < 5\Delta$. On a pass, $|c_s/\widetilde{c}_s| \leq 1 + \Delta/t < 3/2$ and the signs agree. On a failure, $|c_s| < 6\Delta$, including when the estimate is zero. Since future Fourier samples have probabilities c_s^2,

$$h(x) = \sum_{s\in S:\, s \text{ passes}} \frac{c_s^2}{\widetilde{c}_s}\chi_s(x).$$

The coefficient error on a pass is $c_s(c_s - \widetilde{c}_s)/\widetilde{c}_s$, of magnitude at most $3\Delta/2$. On a failure it is at most 6Δ. The passing and failing sets partition S, so Parseval bounds the total squared error by $36|S|\Delta^2 \leq 36\varepsilon^2 e^2$. A single weighted sample has second moment $\sum_{s \text{ passes}} c_s^2/\widetilde{c}_s^2 \leq 9k/4$, independently of x. Independence within each group proves the second inequality. The function h is analysis-only; the algorithm evaluates H, not h. □

Lemma 7 (Bounded majority and original residual scale)**.** *Take $m = 144k$. Conditional on* (12)*,*

$$\mathbb{E}_{\text{groups}}\,\|H - f\|_2^2 \leq 576\varepsilon^2 e^2 + 4\cdot 2^{-g}. \tag{16}$$

If $\varepsilon^2 \leq \beta/262144$ and $2^g \geq 512k^2/\beta$, then for $W > 0$,

$$\Pr_{\text{groups}}\Big[\|H - f\|_2^2 > W/64\Big] \leq \beta. \tag{17}$$

Proof. Let $A = \{x : |h(x) - f(x)| > 1/2\}$. Its measure is at most $4\,\|h - f\|_2^2$. Outside A, h has the correct sign with margin at least $1/2$. Lemma 6 and Chebyshev bound each group's sign-error probability there by $1/16$. For a fixed such x, the groups are independent conditional on the estimator batch. Summing over possible erroneous majorities bounds their probability by $2^g(1/16)^{g/2} = 2^{-g}$. No independence over different inputs x is asserted. Integrating, and using the squared error bound four for Boolean outputs, gives

$$\mathbb{E}_{\text{groups}}\,\|H - f\|_2^2 \leq 16\,\|h - f\|_2^2 + 4\cdot 2^{-g},$$

which proves (16). Since $e^2 \leq \alpha^2 W$ and $\alpha^2 < 2$, the first term is at most $\beta W/128$. By (3), positive W is at least $1/k^2$, so the second term is also at most $\beta W/128$. Markov gives (17). □

The majority in (14) is taken after taking signs. Bounded output controls the exceptional set A even if the rational averages are large there. This argument does not assume an unbiased computed hypothesis or stability of an unbounded majority polynomial.

5 Complete algorithm and success probability

The generic lemmas above allow their failure parameters to be chosen separately. Use constant proposal confidence ρ and smaller archive-safety failure budget β:

$$\begin{aligned} \ell &= \lceil \log_2(2k) \rceil, & T &= 8\ell, & \beta &= \frac{1}{200T}, \\ \rho &= \frac{1}{16}, & \varepsilon &= \frac{1}{2048}, & \gamma &= \frac{1}{32}, \\ m &= 144k, & \tau &= \frac{1}{4k}, & P &= 64k^2. \end{aligned} \tag{18}$$

Choose the least positive odd g such that $2^g \geq 512k^2/\rho = 8192k^2$, and set

$$\begin{aligned} q_e &= \left\lceil 32k\varepsilon^{-2} \ln \frac{4(k+1)}{\rho} \right\rceil, \\ q_c &= \max\left\{ \left\lceil 32k\gamma^{-2} \ln \frac{4(P+1)}{\beta} \right\rceil, \left\lceil 300(k+1) \ln \frac{2(k+1)}{\beta} \right\rceil \right\}. \end{aligned} \tag{19}$$

For an exact finite implementation, replace each $\ln z$ in a sample count by $\lceil \log_2 z \rceil$. This only increases the count and is computed by integer comparisons. The same convention applies to search and final validation. In Lemma 2, the proposal uses list-size bound k and failure budget ρ, while certification uses P, accuracy γ and failure budget β. The generic failure parameter in Lemmas 7 and 5 is instantiated as ρ.

Initialize $U = \varnothing$ and $a = 0$. Execute exactly T trials unless an explicitly stated global abort returns FAIL:

1. Measure q_e fresh labels. Store the residuals and v and define (11). These estimates precede the new Fourier samples.

2. Fourier-measure $4gm$ new copies, discarding every flag-zero outcome. If fewer than gm flags succeed, set $L = \varnothing$ and proceed to step 4. Otherwise split the first gm successful indices into g groups. If more than k distinct grouped indices occur, again set $L = \varnothing$ and proceed to step 4. This distinct-count guard cannot fire for ideal examples of an admitted target. Compute the estimates at these indices and construct (14).

3. Apply Lemma 5 to this known H with τ, P and failure budget ρ. Use its returned list, or set $L = \varnothing$ if the search returns FAIL. A failed proposal does not abort the learner or modify its archive.

4. Measure q_c new labels, including when L is empty. Apply (7) to $L \setminus U$ with parameter γ, using the old residual $f - a$. Add all admissions permanently, including indices whose fitted coefficients may later be zero. Globally abort if $|U| > k$. Refit the original target labels $f(X_j)$ from this same batch exactly on the updated archive; globally abort on a singular nonempty system or $\|a\|_2^2 > 2$. The empty fit is zero.

After the last trial, remove zero fitted coefficients, check sparsity at most k, and verify $a^2 = 1$ by exact Fourier convolution. The convolution has at most k^2 pairs; it does not enumerate the cube. Finally compare a with $\lceil 2k \ln 100 \rceil$ fresh labels. Return the exact list if all checks pass, and otherwise return FAIL. No stopping decision uses the unknown energy W or an inference from sampled $v = 0$.

Proposition 8 (Safety for arbitrary proposals)**.** *Condition on any history with $U \subseteq S$ and* (9)*. For every proposal list L of size at most P fixed before step 4, let C be the residual accuracy event of Lemma 2 with parameter γ, and let G be the full-support Gram event of Lemma 4. Their combined failure probability is at most 2β. On $C \cap G$, no global abort occurs, the updated state satisfies* (9)*, and*

$$U \subseteq U^+ \subseteq S, \qquad W(U^+) \le W(U).$$

If $W(U) = 0$, the new fit remains exactly f.

Proof. The list precedes the fresh certification batch, even if it depends on all earlier data and an inaccurate H. On C, Lemma 3 admits only true indices. Thus $|U^+| \le k$. On G, the nonsingularity, error and norm guarantees hold for every subset of S, including U^+ chosen using these same rows. The fit therefore satisfies (9) and its squared norm is less than two. No archive index is deleted, so missing original energy cannot increase. When $W = 0$, the old fit is f, all certificate residuals are identically zero and no new index is admitted; the Gram event refits f exactly. The proposal may still contain indices outside S; zero certificate residuals reject these as well. The events C, G need not be independent. Their probabilities are bounded conditionally on each history preceding their shared fresh batch, and a union bound gives 2β. □

Proposition 9 (Constant conditional progress)**.** *For every history satisfying the preceding invariant with $W(U) > 0$, a trial safely preserves that invariant and gives $W(U^+) < W(U)/16$ with probability at least $3/4 - 2\beta \ge 1/2$.*

Proof. Use failure budget ρ in the proposal estimates, majority lemma and prefix search. Since $\varepsilon^2 = \rho/262144$, Lemma 7 gives $\|H - f\|_2^2 \le W/64$ except with conditional probability at most ρ on the estimator event. The flag count is below gm with probability at most $\exp(-gm/4) \le \rho$; conditioning on enough flags preserves the independent Fourier law of the successful indices. A staged union bound over estimates, flags, majority accuracy and search gives simultaneous success with probability at least $1 - 4\rho = 3/4$. No independence among these events is assumed.

On those four events, a true coefficient omitted from L has $|\widehat{H}(s)| < 1/(4k) \le |c_s|/4$, and hence

$$\sum_{s \in S \setminus L} c_s^2 \le \frac{16}{9} \|H - f\|_2^2 \le W/36. \tag{20}$$

On the fresh certification event C, every proposed true frequency still not admitted has magnitude at most $4\gamma e/\sqrt{k}$. Their total energy is at most $16\gamma^2 e^2 \le \alpha^2 W/64$. Therefore

$$W(U^+) \le \left(\frac{1}{36} + \frac{\alpha^2}{64}\right) W < W/16. \tag{21}$$

Intersecting also with G gives a safe contraction. The probability is at least $3/4 - 2\beta \ge 1/2$, because $T \ge 8$ and $\beta \le 1/1600$. All other safety-preserving trials remain nonincreasing in W, whether or not the proposal was accurate. □

Proof of the success claim in Theorem 1. Initially $W = 1$ and (9) holds. Let B be the event that a first safety failure C^c or G^c occurs during the T trials. Before the first such failure the state is valid, regardless of earlier proposal quality. Proposition 8 and a first-failure union bound give

$$\Pr(B) \le 2T\beta = 1/100. \tag{22}$$

We do not condition the entire process on future safety before counting progress.

For the proof only, define indicators $I_1, \dots, I_T$. If a safety failure occurred earlier, or W is already zero at the trial's start, set $I_t = 1$. Otherwise set $I_t = 1$ precisely when the four good proposal events from Proposition 9 and $C \cap G$ occur. If execution terminates, unused indicators are one. A global abort cannot precede a safety failure. Thus, for the filtration $\mathcal{F}_{t-1}$ generated by the past trials, either $I_t = 1$ by convention or the current state is valid with positive W, and

$$\Pr(I_t = 1 \mid \mathcal{F}_{t-1}) \geq 1/2.$$

Put $Z_t = \sum_{j \leq t} I_j$. Conditional expectation yields

$$\mathbb{E}[2^{-I_t} \mid \mathcal{F}_{t-1}] = 1 - \tfrac{1}{2} \Pr(I_t = 1 \mid \mathcal{F}_{t-1}) \leq 3/4, \qquad \mathbb{E}[2^{-Z_T}] \leq (3/4)^T.$$

This iteration does not assume independent trials. By Markov,

$$\Pr(Z_T < \ell) \leq 2^{\ell}(3/4)^T = \left[2(3/4)^8\right]^{\ell} < 1/4, \tag{23}$$

since $\ell \geq 1$ and $2(3/4)^8 = 6561/32768 < 1/4$.

On B^c, zero missing energy is absorbing. If it has not already occurred, $Z_T \geq \ell$ supplies at least ℓ genuine factor-16 contractions, interspersed with nonincreasing safe trials. Consequently

$$W(U_T) < 16^{-\ell} \leq \frac{1}{16k^4} < \frac{1}{k^2}.$$

Granularity forces $W(U_T) = 0$. The final exact refit is therefore f, and every output check passes surely. Combining (22) and (23) gives exact-output probability greater than $1 - 1/100 - 1/4 = 37/50 > 2/3$. Dummy indicators after a safety failure do not imply correctness on that event; its probability is charged separately.

For completeness, the final validation limits erroneous acceptance. If two distinct Boolean functions have sparsity at most k, their difference d has at most $2k$ Fourier terms and $\|d\|_\infty = 2$. Cauchy–Schwarz gives $4 \leq 2k \|d\|_2^2$, so their disagreement probability is at least $1/(2k)$. Conditional on any incorrect candidate passing Booleanity, fresh validation accepts it with probability at most $1/100$. The exact-output success bound above does not depend on this additional error-detection guarantee. □

6 All resource costs and exact arithmetic

The proposal parameters $\rho, \varepsilon, \gamma$ are absolute constants. We have $T, \beta^{-1}, g = O(\log(2k))$ and $q_e, q_c = O(k \log(2k))$. Including every unsuccessful proposal, refit and final validation, the total supplied-copy cap is

$$T(q_e + 4gm + q_c) + \lceil 2k \ln 100 \rceil = O(k \log^2(2k)). \tag{24}$$

The logarithm of n in the prefix-search confidence does not enter this count: those evaluations are of H. Each copy requires at most $O(n)$ single-qubit gates and measurements. No consumed copy is reused and no collective measurement is required.

Refits and coefficient precision. Every refit uses at most k columns of signs and the original sign labels. Its normal-equation Gram matrix and right side are integer, with entries bounded by q_c. Hadamard's determinant bound gives a common denominator and coefficient numerators of bit length

$$b_a = O\big(k \log(2kq_c)\big)$$

on every nonsingular tape, regardless of whether the archive is correct. Fraction-free integer elimination computes the solution in polynomial bit time; singularity returns FAIL. Refitting resets this bit bound each round. No circuit for an earlier H is recursively embedded into the next one.

Estimator residuals share the fit's denominator. Therefore every subsequently evaluated estimate (11) has a common denominator and numerator length $O(b_a + \log(2kq_e))$. The variance estimate has a squared common denominator of comparable polynomial bit length. The cutoff, certification, norm and Booleanity tests use exact rational comparisons, with no numerical approximation.

Evaluating the hypothesis. At most k distinct grouped Fourier indices are stored; multiplicities specify their contributions to each group. The distinct-count cap never fires on valid ideal quantum examples, since each retained flag-one outcome belongs to S; discarded flag-zero outcomes are not counted. Write the nonzero estimates as N_s/D with a common positive integer denominator D. The product of their nonzero $|N_s|$ is a positive common denominator for the inverse weights, after incorporating D in the numerators. Its bit length is

$$O(k[b_a + \log(2kq_e)]) = \widetilde{O}(k^2).$$

Store the resulting signed integer weights and group multiplicities once. To evaluate H, compute the parities and signs of the group sums. Positive common factors and $1/m$ do not affect a sign. This costs $\widetilde{O}(nk + k^3)$ bit operations per query, using schoolbook integer arithmetic. The estimate cutoff prevents an uncontrolled statistical inverse weight; the common-denominator calculation also pays its exact computational representation cost.

Classical time, randomness and space. The search uses at most $2nP = O(nk^2)$ prefix tests. Each takes $O(k^4 \log(nk/\rho))$ samples. Consequently there are $O(nk^6 \log(nk/\rho))$ evaluations of H per round, which give (2) after charging the T rounds. Generating the shared-suffix input bits is included. Computing estimator residuals and requested coefficients, streaming candidate certificates, constructing the final at-most-k Gram matrix and exact elimination, and checking the final convolution have smaller polynomial costs than this conservative bound.

Store the current P prefixes or candidates, label rows for the estimator and certificate, the archive, sampled frequencies with multiplicities, integer weights, and streamed candidate accumulators. Candidate accumulators have $\widetilde{O}(k)$ bits each; there are $O(k^2)$ of them. The total is $\widetilde{O}(nk^2 + k^3)$ live bits. After admissions, compute the new Gram matrix from the stored certificate rows; a P-by-P matrix is unnecessary. All temporary integer arithmetic fits these bounds. Old hypotheses and batches are discarded once the new fit is available.

Every trial count, batch size, prefix count and arithmetic input size is capped. An unsuccessful proposal yields an empty list; its consumed work and subsequent refit are included. Archive overflow is checked before forming a fit, so its dimension never becomes the larger candidate-list bound. On bad histories the known function H is still Boolean, since it is a sign-majority. Zero inverses, ties, singular systems and archive overflow have explicit rules. Thus every tape terminates within the stated polynomial bounds, not merely every successful tape. This completes Theorem 1.

7 Scope, limitations and verification status

The algorithm bounds both sample use and computation in the noiseless model of supplied examples. It does not improve Ye's logarithmic sample factor [2], achieve the $O(k \log k)$ target, or provide a practical implementation. Its constants and polynomial exponents are deliberately loose. Noise

robustness and improved classical runtime require separate analysis. The bound k on target sparsity is an input promise. If the actual sparsity exceeds k, the finite resource caps still hold, but neither correctness nor detection of that violation is claimed. The final validation guarantee assumes a target of sparsity at most k.

The known H can have degree n and exponentially many Fourier coefficients. This is permitted because its evaluation cost is polynomial and it only proposes indices. The sparse residual used for estimation and certification is always $f - a$. The proof does not require a generic stability theorem for nonlinear corrections of sparse fits, or a short representation of every recovered index in the original samples.

After measurement, the algorithm is classical: its data consist of uniform labeled examples and successful Fourier samples with law c_s^2. The access model does not yield a general classical simulation of quantum computation or decide BPP versus BQP. Quantum-example preparation is not included in the theorem. Comparison with classical random-example bounds must keep this difference in supplied data explicit.

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